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RESEARCH ARTICLE

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US temperatures: Time trends and persistence

Luis A. Gil-Alana[1] | **Laura Sauci**[2]

[1]University of Navarra, Pamplona, Spain

[2]University of Huelva, Huelva, Spain

Correspondence
Luis A. Gil-Alana, Faculty of Economics and NCID-ICS, University of Navarra, Edificio Amigos, E-31009 Pamplona, Spain.
Email: alana@unav.es

Abstract

This paper investigates the time trend coefficients in the temperatures in 48 US states using monthly data from January 1895 to December 2017, as well as in their anomalies with respect to the base period 1901–2000. For this purpose, we use techniques based on fractional integration, which is a more general approach than the standard methods used in the literature based on integer differentiation. The results indicate that with the exception of 10 states, in the remaining 38, the temperature anomalies have increased across time, with the increase being higher than the one expected under the other more standard approaches. The highest increases correspond to New Jersey and Rhode Island, with an increase of approximately 2.9°C over the last 100 years.

KEYWORDS

climate change, fractional integration, global warming, long memory, time trend, surface temperature

1 | INTRODUCTION

There is a general agreement that temperatures have increased over time, not only from a global perspective (Northern and Southern Hemisphere temperatures) but also from a local one, that is, at different locations in time. Both the National Oceanic and Atmospheric Administration (NOAA) and NASA coincide in pointing out that, since 2010, we have had the warmest years on record as compared to the baseline average of the 20th century and of 1951–1980, respectively. In fact, this trend is likely to continue in the long-term due to the continuous greenhouse gas concentrations increase with no clear evidence of a slowdown or "pause" of the global warming rate (Hansen *et al.,* 2010; Cahill *et al.,* 2015).

A common way to assess this problem is by diagnosis of the deterministic or stochastic nature of the trend term in temperature time series. In this context, the empirical literature provides inconclusive evidence. Some authors find a positive, deterministic and statistically significant trend (Bloomfield and Nychka, 1992; Galbraith and Green, 1992; Zheng and Basher, 1999; Fomby and Vogelsang, 2002; Vogelsang and Franses, 2005; Fatichi *et al.,* 2009; Gay *et al.,* 2009). Others, by contrast, observe stochastic features which may suggest the intrinsic nature of climate changes and thus, certain doubts about the warming effects induced by human action (Gordon, 1991; Woodward and Gray, 1993, 1995; Kärner, 1996; Stern and Kaufmann, 2000; Kaufmann *et al.,* 2006, 2010; Ivanov and Evtimov, 2010; Lai and Yoon, 2017). In addition, empirical evidence seems to support that stochastic trends are better represented by long-range dependent or fractionally integrated models (Bloomfield, 1992; Koscielny-Bunde *et al.,* 1998; Percival *et al.,* 2001; Monetti *et al.,* 2003; Gil-Alana, 2005, 2008, 2015; Rybski *et al.,* 2006, 2008; Fatichi *et al.,* 2009; Franzke, 2010, 2012; Bunde *et al.,* 2014; Yuan *et al.,* 2015; Ludescher *et al.,* 2016; Bunde, 2017).

Accordingly, and on a global scale, we can cite for instance the study of Gil-Alana (2005) based on a fractional integration analysis of the Northern Hemisphere temperatures. Using monthly seasonally adjusted data from 1854 to 1999, the author finds that the time series is highly persistent and exhibits long-memory behaviour. In a later research, Gil-Alana (2008) investigates the global and hemispheric anomaly temperatures using fractional integration techniques again but this time with segmented trends from both

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parametric and non-parametric approaches (Bloomfield, 1973). In that study, the author finds out that all the series are long memory with statistically significant time trends but the results are different depending on the series under consideration and the model being estimated. From a non-parametric approach and assuming the existence of a single break in the data, the evidence shows that the regime shift takes place at 1964, for the global and Southern Hemisphere temperatures and, at 1971, for the Northern Hemisphere. That implies an increase of 2.37°C/100 years in the Northern Hemisphere and a 1.30°C for the southern and global temperatures, after that break-point date. For the case of no breaks, the results point to an increase of temperatures ranging between 0.46°C and 0.51°C over the last 100 years. These results are also consistent with Gil-Alana (2015) for the sea surface temperature data. In relation to the structural breaks, there is evidence that suggests that the identification of a single or multiple break can indicate that the climate not only can shift gradually but also abruptly (Hare and Mantua, 2000; Alley *et al.,* 2003; Ivanov and Evtimov, 2010; Coggin, 2012).

At a local scale, there exist various studies addressing the effects of the climate variability taking into account the long-memory properties of time series. This is the case of Gil-Alana (2003), for the study of the monthly Central England Temperature (CET) series; Caballero et al. (2002), for the multidecadal daily time series from CET, Chicago and Los Angeles; Gil-Alana (2010), for the monthly mean, maximum and minimum temperatures in specific locations in Spain and Portugal; Franzke (2012), for the daily temperatures from four stations: central England, Stockholm (Sweden), Faraday-Vernadsky (Antarctic Peninsula) and Alert (Canada), Percival *et al*. (2001) and Gil-Alana (2012), for monthly temperatures in Alaska; Kane and Usman (2013) for the Sokoto metropolits in Nigeria, or Franzke (2010), Bunde *et al*. (2014), and Ludescher *et al*. (2016), for monthly temperatures of different stations of Antarctica.

Regarding temperature trends in the contiguous 48 United States, we could cite, for instance, the work of Lu *et al*. (2005), as an update of the analysis addressed by Lund et al. (2001). In it, the authors apply a parsimonious changepoint parameterization technique on a US Historical Climatology Network (USHCN) dataset, for the period from January 1812/1926 to December of 2000. Their result shows that winter is the season with the largest variability and with more warming. In general, their results point that the average annual trend estimate is 0.823°C century^{-1} at all 969 stations studied. Thus, and compared to Lund et al. (2001), where the average annual trend was 0.725°C century^{-1}, the last 4 years incorporated into the study seem to have contributed to a more warming. Other example is the research of Capparelli *et al*. (2013). Using the same data source (USHCN) over the period 1898–2008, the authors asses 1,167 station temperature records, by employing a nonlinear spatiotemporal analysis. Their empirical results show strong evidence in favour of warming trend in the contiguous US, except in the South-East area. Finally, and using a fractional approach, we can cite the work of Gil-Alana (2018) who examines the time trends of the maximum and minimum global temperatures–and the differences between them–using annual data from 1895 to 2017. His empirical results support again the hypothesis of global warming.

This paper deals with the analysis of the temperatures and their anomalies across the US states by testing if there is an increase in the values across time and allowing for fractional integration or long memory processes. For this purpose, we estimate a time trend, testing if the time trend coefficient is statistically significantly positive. However, instead of imposing that the associated errors are integrated of order 0 ($I[0]$) as is standard in the literature (see, previous works above mentioned), we estimate the order of integration from a fractional viewpoint. In other words, the errors may be integrated of order d, or $I[d]$, where the differencing parameter d may be a value constrained between 0 and 1, and this value is estimated from the data rather than imposed to be 0. Thus, we start by testing if the US state temperatures and their anomalies are short memory (i.e, $I[0]$) versus the alternative hypothesis of long memory or fractionally integrated (i.e., $I(d)$ with $d > 0$) and, in case of finding support for the alternative, we estimate the time trend coefficient under the proper degree of integration for each series. This is important noting that the estimation of the time trend coefficients may be seriously biased under a wrong specification in the order of integration of the series.

The structure of this article is as follows. Section 2 describes the methodology used and the models employed in the paper. Section 3 presents the data and the main empirical results, while Section 4 concludes the paper.

2 | METHODOLOGY AND MODELS

To test if the temperatures have increased over time, the usual approach consists of using the following model

$$y_t = \beta_0 + \beta_1 t + x_t; \quad t = 0, 1, \ldots, \tag{1}$$

where y_t is the temperature data (or its anomalies with respect to a certain period), β_0 and β_1 are unknown coefficients referring respectively to the intercept and the time trend, and x_t is the error term that is supposed to be $I[0]$ as defined below. In this context, the parameter β_1 measures the average change in y per time period. Thus, climate warming may be occurring if β_1 is positive, in which case there is an increasing trend in the temperatures. From a

statistical viewpoint, we claim that temperatures are increasing if we are able to reject the null hypothesis

$$H_o = \beta_1 = 0 \tag{2}$$

In Equation (1) against the one-sided alternative: $\beta_1 > 0$.

However, a crucial point here is the estimation of β_1 in Equation (1), where the error term is supposed to satisfy some statistical properties. Thus, for example, if x_t in Equation (1) is a random variable independently drawn from a Gaussian distribution with zero mean and constant variance, the OLS estimate of β_1 can be efficiently calculated, and inference may be made based on the standard F and t statistics. (see, Hamilton, 1994, Chapter 16). In a more general sense, the classical estimation based on ordinary (OLS) or generalized (GLS) least squares requires the assumption that x_t in Equation (1) is "well-behaved," in the sense that if there is any dependence between the errors, it is of a "weak-form."[1] In other words, the error term must be integrated of order 0 ($x_t \approx I(0)$), which is an assumption that, as previously argued, might not be satisfied by climatological data.[2] In fact, it is now widely accepted that temperature time series data may display long memory and fractional integration behaviour (see, Vyushin *et al.,* 2009, 2012; Gil-Alana, 2015, 2017, and the references presented above). Thus, conducting the analysis on the regression model (1) under the wrong assumption of $I(0)$ behaviour for the error term may produce a biased estimate of the time trend coefficient.

We first define the concept of long memory that was first pronounced by Hurst (1951).[3] Given a zero-mean covariance stationary process $\{x_t, t = 0, \pm1, \ldots\}$ with autocovariance function $E[(x_t - Ex_t)(x_{t-j} - Ex_t)] = \gamma_j$, we say that x_t displays the property of long memory if:

$$\lim_{T\to\infty} \sum_{j=-T}^{T} |\gamma_j| = \infty \tag{3}$$

Similarly, and using the frequency domain counterpart, assuming that x_t has an absolutely continuous spectral distribution function, with a spectral density function given by…

$$f(\lambda) = \frac{1}{2\pi} \sum_{j=-\infty}^{\infty} \gamma_j \cos\lambda_j \quad -\pi < \varepsilon \leq \pi \tag{4}$$

long memory takes place if the spectral density function of x_t is unbounded at some frequency λ in the interval $[0, \pi]$, i.e.,

$$f(\lambda) \to \infty, \quad \text{as } \lambda \to \lambda, \quad \lambda^* \in [0, \pi] \tag{5}$$

(see McLeod and Hipel, 1978) In this context, a very typical model satisfying the above two properties is the one based on the concept of fractional integration.

We say that a process $\{x_t, t = 0, \pm1, \ldots\}$ is (fractionally) integrated of order d, and denoted as $x_t \approx I(d)$, if it can be represented as

$$(1-B)^d x_t = u_t, \quad t = 0, \pm1, \ldots, \tag{6}$$

where B refers to the backshift operator, that is, $Bx_t = x_{t-1}$, d can be a fractional value and u_t is supposed to be $I(0)$ as defined in footnote number 2. Examples of $I(0)$ processes are the white noise case and the stationary and invertible AutoRegressive Moving Average (ARMA) processes. Papers using fractional integration and long memory in climatological data are Montanari *et al.* (1996), Stephenson *et al.* (2000) Percival *et al.* (2001), Gil-Alana (2003, 2005, 2006, 2008, 2017), Vyushin and Kushner (2009), Zhu *et al.* (2010), Lennartz and Bunde (2009), Rea *et al.* (2011), Franzke (2012), Yuan *et al.* (2013), Bunde et al., (2014), Yuan *et al.* (2015), Ludescher *et al.* (2016), Lustig *et al.* (2016), Bunde (2017), Yuan *et al.* (2019), Anh *et al.* (2019), etc.

3 | DATASET AND EMPIRICAL RESULTS

In this paper, we examine the average temperatures and their anomalies (measured in degrees Celsius) in 48 US states. The two series have monthly frequency and cover the period January 1895–December 2017, with the base period being 1901–2000. The data source is Vose *et al.* (2014) and it has been collected from the NOAA's National Centers for Environmental Information (www.ncdc.noaa.gov).

We start by considering the following model

$$y_t = \beta_0 + \beta_1 t + x_t, \quad (1-B)^d x_t = u_t, \quad t = 1, 2, \ldots, \tag{7}$$

where y_t is the observed time series data; β_0 and β_1 are unknown coefficients corresponding to an intercept and a linear time trend, d is the (potentially) fractional differencing parameter, and based on the monthly structure of the data, we suppose u_t follows a seasonal (monthly) AR(1) process of the following form

$$u_t = \varphi u_{t-12} + \varepsilon_t, \quad t = 1, 2, \ldots \tag{8}$$

where φ refers to the seasonal AR coefficient and ε_t is a white noise process. Thus, φ indicates the degree of seasonal dependence. This specification makes sense noting that β_1 determines the existence or not of a linear time trend;

TABLE 1 Estimated parameters in the model given by Equation (1)

State	d – Diff. parameter	α (Intercept)	β (Time trend)	φ Seasonality
Alabama	0.15 (0.09, 0.21)	63.022 (42.97)	0.00016 (0.09)	0.941
Arizona	0.18 (0.13, 0.24)	58.299 (32.17)	0.00182 (0.89)	0.963
Arkansas	0.10 (0.04, 0.15)	60.213 (46.36)	0.00041 (0.28)	0.950
California	0.15 (0.10, 0.21)	56.163 (43.07)	0.00196 (1.33)	0.951
Colorado	0.13 (0.08, 0.18)	43.353 (27.58)	0.00196 (1.10)	0.962
Connecticut	0.17 (0.13, 0.23)	46.374 (23.10)	0.00263 (1.12)	0.961
Delaware	0.14 (0.09, 0.20)	52.838 (32.27)	0.00220 (1.19)	0.957
Florida	0.21 (0.15, 0.28)	69.150 (50.87)	0.00149 (0.97)	0.919
Georgia	0.15 (0.10, 0.22)	63.017 (44.40)	0.00061 (0.38)	0.944
Idaho	0.17 (0.12, 0.23)	32.341 (42.71)	0.00153 (0.73)	0.946
Illinois	0.10 (0.05, 0.16)	50.838 (32.40)	0.00124 (0.69)	0.950
Indiana	0.11 (0.04, 0.17)	50.670 (31.87)	0.00108 (0.59)	0.948
Iowa	0.11 (0.07, 0.17)	46.748 (24.36)	0.00116 (0.53)	0.951
Kansas	0.05 (0.01, 0.11)	53.374 (44.29)	0.00129 (0.88)	0.956
Kentucky	0.10 (0.04, 0.16)	55.228 (40.57)	0.00052 (0.33)	0.946
Louisiana	0.16 (0.11, 0.22)	66.070 (46.02)	0.00054 (0.33)	0.939
Maine	0.18 (0.12, 0.23)	38.362 (16.80)	0.00256 (0.95)	0.966
Maryland	0.13 (0.08, 0.19)	52.633 (33.40)	0.00183 (1.02)	0.957
Massachusetts	0.17 (0.12, 0.23)	45.377 (22.52)	0.00239 (1.05)	0.962
Michigan	0.19 (0.14, 0.25)	41.875 (17.04)	0.00245 (0.78)	0.954
Minnesota	0.16 (0.11, 0.22)	38.450 (14.89)	0.00243 (0.83)	0.953
Mississippi	0.13 (0.08, 0.20)	63.550 (47.12)	0.00020 (0.13)	0.942
Missouri	0.08 (0.03, 0.14)	54.042 (39.93)	0.00078 (0.50)	0.950
Montana	0.09 (0.04, 0.14)	39.796 (27.87)	0.00204 (1.25)	0.932
Nebraska	0.12 (0.08, 0.18)	58.072 (43.96)	0.00088 (0.59)	0.951
Nevada	0.14 (0.08, 0.19)	37.847 (16.02)	0.00257 (0.96)	0.944
New Hampshire	0.17 (0.11, 0.23)	40.823 (18.47)	0.00233 (0.94)	0.962
New Jersey	0.16 (0.11, 0.22)	49.723 (26.22)	0.00268 (1.27)	0.959
New Mexico	0.19 (0.14, 0.24)	51.833 (27.02)	0.00168 (0.78)	0.967
New York	0.15 (0.10, 0.21)	43.404 (22.18)	0.00179 (0.81)	0.957
N. Carolina	0.06 (0.01, 0.11)	47.470 (35.98)	0.00145 (0.95)	0.952
N. Dakota	0.17 (0.12, 0.23)	48.331 (26.37)	0.00185 (0.89)	0.946
Ohio	0.12 (0.07, 0.18)	49.721 (30.72)	0.00119 (0.65)	0.949
Oklahoma	0.07 (0.02, 0.13)	59.327 (48.92)	0.00060 (0.43)	0.957
Oregon	0.15 (0.10, 0.21)	45.089 (33.05)	0.00204 (1.32)	0.939
Pennsylvania	0.14 (0.08, 0.20)	47.241 (27.04)	0.00144 (0.73)	0.955
Rhode Island	0.18 (0.13, 0.23)	46.755 (48.71)	0.00278 (1.22)	0.961
S. Carolina	0.14 (0.08, 0.21)	61.937 (44.04)	0.00081 (0.51)	0.948
S. Dakota	0.09 (0.05, 0.15)	43.625 (44.66)	0.00192 (0.99)	0.946
Tennessee	0.10 (0.05, 0.16)	57.499 (44.82)	0.00040 (0.27)	0.946
Texas	0.16 (0.10, 0.21)	63.880 (40.24)	0.00119 (0.66)	0.956
Utah	0.21 (0.16, 0.26)	45.849 (18.96)	0.00242 (0.88)	0.955
Vermont	0.15 (0.09, 0.21)	39.957 (41.54)	0.00221 (0.96)	0.959

TABLE 1 (Continued)

State	d – Diff. parameter	α (Intercept)	β (Time trend)	φ Seasonality
Virginia	0.11 (0.06, 0.17)	54.198 (40.31)	0.00106 (0.69)	0.954
Washington	0.10 (0.05, 0.16)	51.233 (38.65)	0.00070 (0.46)	0.941
W. Virginia	0.16 (0.11, 0.22)	45.176 (30.63)	0.00144 (0.86)	0.948
Wisconsin	0.16 (0.09, 0.22)	41.205 (17.50)	0.00184 (0.69)	0.953
Wyoming	0.09 (0.04, 0.14)	39.404 (28.70)	0.00207 (1.32)	0.953

The values in parenthesis in the second column refer to the 95% confidence intervals of d; in the third and fourth column, they are t-values.

d indicates the degree of persistence in the data, while φ refers to the seasonal pattern. Of course, other more complex structures can be considered. For instance, nonlinear time trends can be taken into account and other more elaborated (ARMA) models can be utilized in the specification of u_t in (7). However, it seems that a simple model like the one presented in Equations (7) and (8) can be sufficient to describe the main three features of the data, that is, the time trend, the persistence and the seasonal patterns. The estimation is conducted by means of a Whittle procedure in the frequency domain using the methods described in Dahlhaus (1989) and Robinson (1994). The Whittle function is simply an approximation to the likelihood function (Whittle, 1953), and Robinson (1994) proposed a test statistic for testing the null hypothesis of

$$H_o: \quad d = d_o \tag{9}$$

in a model given by the Equations (7) and (8) for any real value d_o, including thus stationary ($d_o < 0.5$) and nonstationary ($d_o \geq 0.5$) cases. This method has several advantages compared with other methods. First, as mentioned above, it is valid even in nonstationary contexts, and its limit distribution is standard Normal, that is, $N(0, 1)$. Moreover, this asymptotic behaviour holds independently of the inclusion or not of deterministic terms like intercepts and/or linear trends and it is the most efficient method in the Pitman sense against local departures from the null.[4]

Table 1 focusses on the observed time series data, that is, the average monthly temperatures in the 48 US states examined. We start with this dataset because we want to verify if the seasonal monthly effect corresponding to the annual cycle is relevant in the data. This feature is captured through the φ coefficient in Equation (8). The first thing we observe in this table is that the estimated values of d are statistically significantly positive in all cases; note that the 95% confidence intervals associated to these values exclude the value of 0 in all cases, rejecting thus, the hypothesis of short memory or $I(0)$ behaviour. The largest values of d are obtained in the cases of Florida and Utah ($d = 0.21$) while the lowest degrees of persistence correspond to Kansas (0.05) followed by North Carolina (0.06) and Oklahoma (0.07). However, the time trend coefficients, though positive in all cases, are found to be statistically insignificant. On the other hand, the seasonal autoregressive coefficient is very close to 1, in fact, higher than 0.94 in all cases investigated, which suggest the nonstationary character of the seasonal component of the series.

Based on the evidence provided by these large coefficients for the seasonal component, we next perform seasonal unit roots (or seasonal $I(1)$) tests, using Dickey Hasza and Fuller (Dickey *et al*., 1984) and Hylleberg, Engle, Granger and Yoo (Hylleberg *et al*., 1990) methods; although not reported, the results supported the existence of seasonal $I(1)$ behaviour in all cases. Because of this, we perform seasonal differences and conduct the estimation of (7) where y_t is now replaced by $(1 - B^{12})y_t$, that is, $y_t - y_{t-12}$.[5] Note however that in the process of seasonal differencing, many information of the data are removed but this is a pre-requisite for statistical inference, based on the nonstationary nature of the data. The results in terms of the estimates of d and the time trend coefficient (β_1) in (7) are reported in Table 2. Here, we examine two cases for the error term. First, we suppose it is uncorrelated and assume a white noise process, and then, we allow for autocorrelation, employing the exponential spectral model of Bloomfield (1973). The latter is a non-parametric model in the sense that u_t is implicitly defined in terms of its spectral density function, with the autocorrelations decaying exponentially fast as in the autoregressive (AR) case.[6] If u_t is uncorrelated, the higher values of d are those of Utah ($d = 0.20$), Arizona, Florida and New Mexico (0.18) and the $I(0)$ hypothesis is rejected in all the states in favour of long memory ($d > 0$) behaviour. If we impose autocorrelation for the error term throughout the model of Bloomfield (1973), the estimates of d are now smaller, and the $I(0)$ hypothesis cannot be rejected in any single case with the exception of New Mexico (0.13). On the other hand, the time trend coefficients are once more statistically insignificant in all cases.[7] This lack of significance, however, does not imply the lack of an increase in the temperatures since the analysis has been conducted on the monthly differenced series and to answer that question the work must be conducted on the anomalies, once the annual cycle has been removed. For this purpose,

TABLE 2 Estimated coefficients in Equation (1) based on the first differenced data

	No autocorrelation		Autocorrelation (Bloomfield)	
State	***d* – Diff. parameter**	**β (Time trend)**	***d* – Diff. parameter**	**β (Time trend)**
Alabama	0.12 (0.07, 0.18)	−0.00004 (−0.65)	−0.09 (−0.18, 0.01)	0.00002 (0.16)
Arizona	0.18 (0.13, 0.24)	0.00005 (0.09)	0.07 (−0.05, 0.18)	0.00007 (0.17)
Arkansas	0.08 (0.03, 0.13)	−0.00002 (−0.04)	−0.07 (−0.16, 0.01)	0.00002 (0.11)
California	0.14 (0.10, 0.19)	0.00004 (0.09)	−0.01 (−0.09, 0.14)	0.00005 (0.28)
Colorado	0.12 (0.07, 0.17)	−0.00001 (−0.02)	0.05 (−0.02, 0.19)	0.00001 (0.04)
Connecticut	0.16 (0.12, 0.22)	−0.00002 (−0.04)	0.06 (−0.05, 0.18)	0.00002 (0.07)
Delaware	0.13 (0.07, 0.18)	0.00001 (0.02)	−0.02 (−0.10, 0.12)	0.00005 (0.22)
Florida	0.18 (0.12, 0.24)	−0.00001 (−0.02)	−0.11 (−0.21, −0.02)	0.00003 (0.33)
Georgia	0.13 (0.07, 0.19)	−0.00002 (−0.04)	−0.11 (−0.21, 0.18)	0.00003 (0.28)
Idaho	0.16 (0.11, 0.21)	−0.00006 (−0.09)	0.08 (−0.03, 0.01)	−0.00001 (−0.04)
Illinois	0.08 (0.03, 0.13)	−0.00001 (−0.02)	−0.06 (−0.15, 0.18)	0.00002 (0.02)
Indiana	0.09 (0.04, 0.14)	−0.00006 (−0.07)	−0.07 (−0.14, 0.01)	0.00005 (0.17)
Iowa	0.10 (0.05, 0.15)	−0.00006 (−0.01)	−0.05 (−0.10, 0.04)	0.00003 (0.12)
Kansas	0.04 (0.00, 0.09)	0.00005 (0.001)	−0.05 (−0.12, 0.03)	0.00001 (0.07)
Kentucky	0.07 (0.02, 0.12)	−0.00002 (−0.05)	−0.12 (−0.21, 0.00)	0.00002 (0.12)
Louisiana	0.13 (0.08, 0.19)	−0.00003 (−0.08)	−0.09 (−0.17, 0.02)	0.00002 (0.20)
Maine	0.16 (0.11, 0.22)	0.00004 (0.06)	−0.01 (−0.12, 0.08)	0.00004 (0.16)
Maryland	0.12 (0.07, 0.17)	−0.00001 (−0.03)	−0.03 (−0.13, 0.07)	0.00003 (0.14)
Massachusetts	0.16 (0.12, 0.21)	0.00001 (0.01)	0.06 (−0.04, 0.10)	0.00003 (0.09)
Michigan	0.17 (0.13, 0.23)	−0.00009 (−0.12)	0.02 (−0.09, 0.10)	0.00009 (0.02)
Minnesota	0.14 (0.10, 0.19)	−0.00003 (−0.04)	−0.10 (−0.18, −0.01)	0.00003 (0.18)
Mississippi	0.11 (0.05, 0.17)	−0.00002 (−0.05)	−0.09 (−0.20, 0.00)	0.00002 (0.18)
Missouri	0.06 (0.02, 0.11)	0.00009 (0.02)	−0.08 (−0.19, 0.02)	0.00003 (0.17)
Montana	0.07 (0.03, 0.12)	−0.00002 (−0.05)	0.01 (−0.08, 0.12)	−0.00004 (−0.03)
Nebraska	0.04 (0.00, 0.09)	0.00002 (0.05)	−0.04 (−0.13, 0.07)	0.00001 (0.06)
Nevada	0.16 (0.11, 0.21)	−0.00001 (−0.01)	0.02 (−0.10, 0.12)	0.00003 (0.10)
New Hampshire	0.15 (0.11, 0.21)	0.00001 (0.03)	0.01 (−0.09, 0.14)	0.00004 (0.15)
New Jersey	0.15 (0.10, 0.20)	−0.00009 (−0.01)	0.04 (−0.06, 0.17)	0.00002 (0.08)
New Mexico	0.18 (0.13, 0.23)	−0.00002 (−0.02)	0.13 (0.02, 0.26)	0.00001 (0.02)
New York	0.14 (0.09, 0.19)	−0.00001 (−0.02)	0.01 (−0.10, 0.12)	0.00002 (0.09)
N. Carolina	0.10 (0.05, 0.16)	−0.00002 (−0.05)	−0.10 (−0.19, 0.01)	0.00002 (0.20)
N. Dakota	0.12 (0.08, 0.18)	−0.00004 (−0.05)	0.01 (−0.10, 0.12)	−0.00006 (−0.02)
Ohio	0.10 (0.05, 0.15)	−0.00001 (−0.01)	−0.07 (−0.16, 0.01)	0.00004 (0.21)
Oklahoma	0.16 (0.00, 0.11)	−0.00001 (−0.04)	−0.05 (−0.14, 0.06)	0.00009 (0.04)
Oregon	0.14 (0.09, 0.19)	−0.00008 (−0.15)	0.03 (−0.07, 0.14)	−0.00003 (−0.13)
Pennsylvania	0.12 (0.05, 0.17)	−0.00003 (−0.05)	−0.03 (−0.11, 0.09)	0.00002 (0.11)
Rhode Island	0.17 (0.12, 0.22)	0.00003 (0.06)	0.07 (−0.04, 0.17)	0.00005 (0.14)
S. Carolina	0.12 (0.05, 0.18)	−0.00001 (−0.02)	−0.00 (−0.19, 0.01)	0.00004 (0.30)
S. Dakota	0.08 (0.04, 0.13)	−0.00002 (−0.03)	−0.01 (−0.09, 0.10)	−0.00002 (−0.01)
Tennessee	0.08 (0.03, 0.13)	−0.00001 (−0.02)	−0.11 (−0.20, −0.02)	0.00003 (0.22)
Texas	0.14 (0.09, 0.19)	−0.00006 (−0.13)	0.05 (−0.05, 0.14)	−0.00001 (−0.04)
Utah	0.20 (0.16, 0.26)	−0.00005 (−0.07)	−0.05 (−0.06, 0.15)	0.00001 (0.04)

TABLE 2 (Continued)

State	No autocorrelation		Autocorrelation (Bloomfield)	
	d – Diff. parameter	*β* (Time trend)	*d* – Diff. parameter	*β* (Time trend)
Vermont	0.14 (0.09, 0.20)	0.00001 (0.02)	0.02 (−0.09, 0.12)	0.00004 (0.12)
Virginia	0.09 (0.04, 0.14)	−0.00001 (−0.03)	−0.09 (−0.18, 0.00)	0.00003 (0.18)
Washington	0.15 (0.10, 0.20)	−0.00005 (−0.10)	0.07 (−0.02, 0.16)	−0.00003 (−0.09)
W. Virginia	0.08 (0.03, 0.13)	−0.00007 (−0.01)	−0.09 (−0.18, 0.01)	0.00004 (0.25)
Wisconsin	0.14 (0.09, 0.19)	−0.00005 (−0.07)	−0.01 (−0.10, 0.09)	0.00003 (0.09)
Wyoming	0.08 (0.04, 0.13)	−0.00005 (−0.11)	0.05 (−0.05, 0.14)	−0.00004 (−0.10)

The values in parenthesis in the second and fourth columns refer to the 95% confidence intervals of *d*; in the third and fifth column, they are *t*-values.

we use now the base period 1901–2000, and the results, for the same model as above are presented in Table 3. Again, we consider here the two cases of uncorrelated and autocorrelated errors. For comparisons with the classical model, we also report in Table 3 the time trend coefficients estimated under the assumption that $d = 0$ in (7).

TABLE 3 Estimated coefficients in the anomaly series

States	*d* estimated from the data				*d* = 0
	White noise		Autocorrelation		
	d (Diff. parameter)	*β* (Time trend)	*d* (Diff. parameter)	*β* (Time trend)	*β* (Time trend)
Alabama	0.14 (0.10, 0.18)	0.000003 (0.07)	0.09 (0.04, 0.14)	−0.000006 (−0.21)	−0.00015 (−0.85)
Arizona	0.17 (0.14, 0.22)	**0.001609 (4.35)**	0.15 (0.10, 0.22)	**0.00160 (4.81)**	**0.00156 (10.45)**
Arkansas	0.12 (0.09, 0.16)	0.000329 (0.89)	0.10 (0.05, 0.16)	0.00031 (0.93)	0.00022 (1.17)
California	0.16 (0.12, 0.21)	**0.001824 (5.37)**	0.13 (0.08, 0.20)	**0.00181 (6.22)**	**0.00177 (12.26)**
Colorado	0.13 (0.09, 0.17)	**0.001812 (5.18)**	0.13 (0.07, 0.19)	**0.00181 (5.14)**	**0.00173 (9.98)**
Connecticut	0.16 (0.12, 0.21)	**0.002237 (5.31)**	0.12 (0.06, 0.20)	**0.00221 (6.45)**	**0.00215 (12.05)**
Delaware	0.15 (0.11, 0.19)	**0.002213 (4.91)**	0.12 (0.07, 0.18)	**0.00199 (5.67)**	**0.00192 (10.47)**
Florida	0.18 (0.13, 0.22)	**0.00126 (3.31)**	0.06 (0.01, 0.13)	**0.00118 (5.95)**	**0.00115 (7.87)**
Georgia	0.14 (0.11, 0.19)	0.00046 (1.26)	0.07 (0.03, 0.14)	0.00039 (1.58)	**0.00034 (1.96)**
Idaho	0.16 (0.13, 0.21)	**0.00132 (2.88)**	0.12 (0.06, 0.19)	**0.00130 (3.53)**	**0.00126 (6.50)**
Illinois	0.12 (0.09, 0.17)	**0.00092 (2.13)**	0.07 (0.02, 0.13)	**0.00088 (2.68)**	**0.00083 (3.69)**
Indiana	0.12 (0.08, 0.16)	**0.00110 (2.50)**	0.08 (0.03, 0.13)	**0.00106 (2.99)**	**0.00100 (4.32)**
Iowa	0.12 (0.09, 0.17)	**0.00092 (2.13)**	0.07 (0.02, 0.13)	**−0.00088 (2.68)**	**0.00083 (3.69)**
Kansas	0.14 (0.10, 0.18)	**0.00098 (1.84)**	0.08 (0.03, 0.14)	**0.00092 (2.39)**	**0.00085 (3.39)**
Kentucky	0.10 (0.06, 0.14)	**0.000117 (3.08)**	0.10 (0.03, 0.16)	**0.00117 (3.07)**	**0.00109 (4.95)**
Kentucky	0.12 (0.09, 0.16)	0.00042 (1.05)	0.08 (0.04, 0.13)	0.00037 (1.15)	0.00029 (1.39)
Louisiana	0.16 (0.12, 0.20)	0.00039 (0.98)	0.09 (0.04, 0.15)	0.00031 (1.13)	0.00022 (1.33)
Maine	0.17 (0.13, 0.22)	**0.00219 (4.76)**	0.09 (0.04, 0.16)	**0.00218 (7.25)**	**0.00216 (11.66)**
Maryland	0.14 (0.10, 0.18)	**0.00166 (4.25)**	0.10 (0.05, 0.17)	**0.00163 (5.16)**	**0.00157 (8.51)**
Massachusetts	0.16 (0.12, 0.21)	**0.00209 (5.00)**	0.13 (0.08, 0.20)	**0.00208 (5.80)**	**0.00204 (11.45)**
Michigan	0.18 (0.15, 0.23)	**0.00204 (3.70)**	0.11 (0.06, 0.19)	**0.00200 (5.27)**	**0.00195 (9.23)**
Minnesota	0.17 (0.13, 0.22)	**0.00210 (3.18)**	0.11 (0.05, 0.18)	**0.00206 (4.30)**	**0.00200 (7.52)**
Mississippi	0.14 (0.10, 0.18)	**0.000009 (0.24)**	0.09 (0.05, 0.14)	0.00003 (0.10)	−0.00006 (−0.33)
Missouri	0.11 (0.07, 0.15)	0.00069 (1.69)	0.07 (0.02, 0.13)	0.00065 (1.78)	**0.00059 (2.64)**

TABLE 3 (Continued)

	d estimated from the data				
	White noise		Autocorrelation		$d = 0$
States	*d* (Diff. parameter)	β (Time trend)	*d* (Diff. parameter)	β (Time trend)	β (Time trend)
Montana	0.11 (0.07, 0.15)	**0.00192 (4.05)**	0.09 (0.03, 0.16)	**0.00191 (4.46)**	**0.00185 (7.07)**
N. Carolina	0.13 (0.08, 0.18)	**0.00077 (2.17)**	0.09 (0.05, 0.15)	**0.00073 (2.54)**	**0.00065 (3.67)**
N. Dakota	0.15 (0.11, 0.20)	**0.00231 (3.44)**	0.10 (0.04, 0.18)	**0.00225 (4.37)**	**0.00216 (7.18)**
New Hampshire	0.16 (0.12, 0.20)	**0.00201 (4.50)**	0.12 (0.06, 0.17)	**0.00200 (5.53)**	**0.00198 (10.42)**
New Jersey	0.16 (0.12, 0.20)	**0.00243 (5.72)**	0.11 (0.06, 0.18)	**0.00240 (7.31)**	**0.00235 (13.02)**
New Mexico	0.17 (0.13, 0.21)	**0.00145 (4.12)**	0.18 (0.13, 0.26)	**0.00145 (3.90)**	**0.00134 (9.42)**
New York	0.15 (0.11, 0.20)	**0.00152 (3.37)**	0.12 (0.05, 0.17)	**0.00151 (3.92)**	**0.00146 (7.24)**
Nebraska	0.10 (0.06, 0.14)	**0.00138 (3.35)**	0.08 (0.04, 0.14)	**0.00136 (3.67)**	**0.00130 (5.44)**
Nevada	0.15 (0.11, 0.20)	**0.00163 (3.89)**	0.10 (0.04, 0.15)	**0.00161 (5.01)**	**0.00156 (8.35)**
Ohio	0.13 (0.09, 0.17)	**0.00103 (2.34)**	0.07 (0.02, 0.13)	**0.00098 (3.10)**	**0.00094 (4.31)**
Oklahoma	0.11 (0.07, 0.15)	0.00055 (1.53)	0.10 (0.05, 0.17)	0.00054 (1.58)	**0.00044 (2.21)**
Oregon	0.15 (0.11, 0.19)	**0.00189 (5.01)**	0.11 (0.05, 0.18)	**0.00187 (6.09)**	**0.00182 (10.82)**
Pennsylvania	0.14 (0.10, 0.18)	**0.00123 (2.93)**	0.10 (0.05, 0.17)	**0.00119 (3.54)**	**0.00113 (5.73)**
Rhode	0.17 (0.13, 0.22)	**0.00247 (5.85)**	0.14 (0.08, 0.20)	**0.00246 (6.78)**	**0.00244 (14.35)**
S. Caroline	0.14 (0.11, 0.18)	**0.00068 (1.94)**	0.08 (0.04, 0.14)	**0.00062 (2.32)**	**0.00055 (3.17)**
S. Dakota	0.13 (0.09, 0.17)	**0.00180 (3.23)**	0.10 (0.04, 0.17)	**0.00177 (3.72)**	**0.00168 (6.10)**
Tennessee	0.12 (0.07, 0.16)	0.00032 (0.84)	0.07 (0.03, 0.12)	0.00026 (0.91)	0.00020 (1.00)
Texas	0.16 (0.12, 0.22)	**0.00103 (2.73)**	0.15 (0.09, 0.23)	**0.00102 (2.84)**	**0.00088 (5.46)**
Utah	0.18 (0.14, 0.23)	**0.00206 (4.25)**	0.10 (0.04, 0.18)	**0.00198 (6.28)**	**0.00191 (10.32)**
Vermont	0.15 (0.11, 0.20)	**0.00193 (4.23)**	0.10 (0.05, 0.17)	**0.00192 (5.50)**	**0.00190 (9.32)**
Virginia	0.12 (0.09, 0.17)	**0.00094 (2.69)**	0.09 (0.04, 0.14)	**0.00092 (3.08)**	**0.00085 (4.65)**
Washington	0.17 (0.13, 0.21)	**0.00128 (2.99)**	0.14 (0.08, 0.22)	**0.00128 (3.50)**	**0.00125 (7.23)**
W. Virginia	0.11 (0.07, 0.16)	0.00059 (1.59)	0.08 (0.03, 0.13)	0.00056 (1.80)	**0.00050 (2.49)**
Wisconsin	0.17 (0.13, 0.22)	**0.00153 (2.55)**	0.11 (0.04, 0.16)	**0.00150 (3.46)**	**0.00147 (6.06)**
Wyoming	0.11 (0.07, 0.15)	**0.00196 (5.18)**	0.12 (0.06, 0.17)	**0.00197 (4.89)**	**0.00188 (9.02)**

In bold the statistical coefficients at the 5% level. The values in parenthesis in the second and fourth column refer to the 95% confidence intervals of *d*; in the other column, they are *t*-values.

The first thing we observe in this table is that the estimated values of d are significantly positive in all cases with both uncorrelated and autocorrelated errors. Under the assumption that the errors are uncorrelated, the largest values of d correspond now to Florida, Michigan and Utah (0.81) with white noise errors and New Mexico (0.18) and Arizona (0.15) with autocorrelation. At the other extreme, Kentucky and Nebraska (0.10) with white noise and Florida (0.06) under autocorrelation display the lowest values of d or the lowest degrees of persistence.

Next, we examine the time trend coefficients. We observe six states where the time trend is insignificantly different from zero for the three model specifications, that is, estimating d with uncorrelated and autocorrelated errors, and imposing a priori that $d = 0$. These states are Alabama, Arkansas, Kentucky, Louisiana, Mississippi and Tennessee. Also, it is of interest to see that for another group of four states (Georgia, Missouri, Oklahoma and West Virginia) the time trend is statistically significant assuming $d = 0$ but becomes insignificantly different from 0 when d is estimated rather than imposed to be 0. That means that in these cases the standard way of testing the increase in the temperatures produces upward biases in favour of the warming in the temperatures. For the remaining 38 states, the time trend coefficients are statistically significantly positive across the three models and, interestingly, the time trends coefficients are higher when the value of d is estimated rather than imposed to be 0. The difference in the value of β_1 is something to be expected noting that if we impose $d = 0$, the time trend is estimated throughout the model:

$$y_t = \beta_0 + \beta_1 t + u_t; \quad t = 0, 1, \ldots. \tag{10}$$

However, if we allow x_t to be fractionally integrated or $I(d)$, the model becomes (7), i.e.,

$$y_t = \beta_0 + \beta_1 t + x_t, \quad (1-L)^d x_t = u_t, \quad t = 1, 2, \ldots$$

which can be expressed in terms of a single equation as

$$y^*_t = \beta_0 1^*_t + \beta_1 t^*_t + u_t; \quad t = 1, 2, \ldots. \tag{11}$$

where $y^*_t = (1-L)^d y_t$; $1^*_t = (1-L)^d 1_t$, and $t^*_t = (1-L)^d t$.

Clearly, the estimation of β_1 in Equations (10) and (11) will be different unless $d = 0$, and as we depart from 0, larger differences should be expected.

Table 4 summarizes the increases in the averaged temperature anomalies over 100 years for the 38 states where we observe significant time trend coefficients, observing an overall increase in the values when d is estimated rather than imposed to be 0, being also slightly higher the values under uncorrelated errors.

Figure 1 illustrates the results obtained in Table 4 when d is estimated with autocorrelated errors. The warming trend seems obvious for the majority of cases, in particular for the states of the Northeast Region of United States. In general, the increase in the temperatures is between 1.5 and 2.5°C over 100 years and the highest coefficients correspond to New Jersey and Rhode Island, with values around an increase of 2.9°C. Only in two cases, North Dakota and South Carolina, is an increase of less than 1.0°C observed. There are 10 states (the majority of them located in the Southeast) where there is no evidence of an increase in the temperatures. This evidence is consistent with previous studies such as Lu *et al.* (2005) and Capparelli *et al.* (2013), where it is found significant warming trends in the contiguous U.S except in the South-East. The cooling trend in the South-East, so-called "warming hole" (Pan *et al.,* 2004; Kunkel *et al.,* 2006), is a phenomenon detected in the second half of the twentieth century that may be associated to multiple factors. The results of simulations from Coupled Model Intercomparison Project, phases 3 and 5, -CMIP3, CMIP5-carried out by some authors, suggest that the warming hole could be correlated with the interdecadal Pacific oscillation (Meehl *et al.,* 2012) and the Atlantic multidecadal oscillation in sea surface temperature (Kumar *et al.,* 2013).

4 | CONCLUSIONS

In this paper, we have examined the monthly temperatures and their anomalies (base, 1901–2000) in 48 U.S states for the time period 1895 month 1–2017 month 12. Our main

TABLE 4 Degrees of increase in the temperature anomalies under different scenarios

States	Temperature change (d estimated)		Temp. change
	White noise	Autocorrelation	$d = 0$
Arizona	1.9308°C	1.9200°C	1.8720°C
California	2.1888°C	2.1720°C	2.1240°C
Colorado	2.1744°C	2.1720°C	2.0760°C
Connecticut	2.6844°C	2.6520°C	2.5800°C
Delaware	2.6556°C	2.3880°C	2.3040°C
Florida	1.5120°C	1.4160°C	1.3800°C
Idaho	1.5840°C	1.5600°C	1.5120°C
Illinois	1.3200°C	1.2720°C	1.2000°C
Indiana	1.1040°C	1.0560°C	0.9960°C
Iowa	1.1040°C	1.0560°C	0.9960°C
Kansas	1.1760°C	1.1040°C	1.0200°C
Maine	2.6280°C	2.6160°C	2.5920°C
Maryland	1.9920°C	1.9560°C	1.8840°C
Massachusetts	2.5080°C	2.4960°C	2.4480°C
Michigan	2.4480°C	2.4000°C	2.3400°C
Minnesota	2.2520°C	2.4720°C	2.4000°C
Montana	2.3040°C	2.2920°C	2.2200°C
Nebraska	1.6560°C	1.6320°C	1.5600°C
Nevada	1.9560°C	1.9320°C	1.8720°C
New Hampshire	2.4120°C	2.4000°C	2.3760°C
New Jersey	**2.9160°C**	**2.8800°C**	**2.8200°C**
New Mexico	1.7400°C	1.7400°C	1.6080°C
New York	1.8240°C	1.8120°C	1.7520°C
N. Carolina	2.7720°C	2.7000°C	2.5920°C
N. Dakota	0.9240°C	0.8760°C	0.8700°C
Ohio	1.2360°C	1.1760°C	1.1280°C
Oregon	2.2680°C	2.2440°C	2.1840°C
Pennsylvania	1.4760°C	1.4280°C	1.3560°C
Rhode Island	**2.9640°C**	**2.9520°C**	**2.9280°C**
S. Carolina	0.8160°C	0.7440°C	0.6600°C
S. Dakota	2.1600°C	2.1240°C	2.0160°C
Texas	1.2360°C	1.2440°C	1.0560°C
Utah	2.4720°C	2.3760°C	2.2920°C
Vermont	2.3160°C	2.3040°C	2.2800°C
Virginia	1.1280°C	1.1040°C	1.0200°C
Washington	1.5360°C	1.5360°C	1.5000°C
Wisconsin	1.8360°C	1.8000°C	1.7640°C
Wyoming	2.3520°C	2.3640°C	2.2560°C

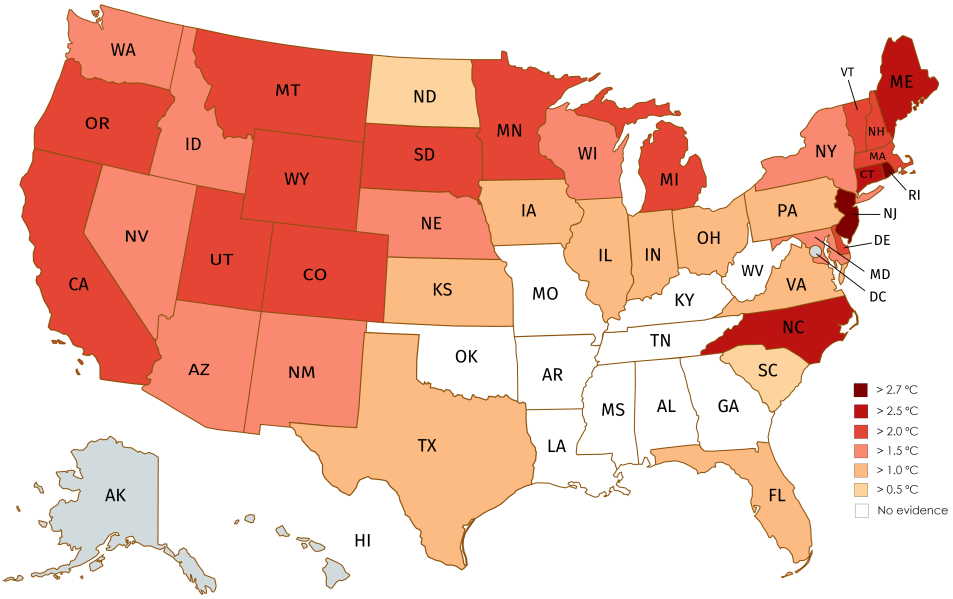


FIGURE 1 Degrees of increase in the averaged temperature anomalies for 38 US states over 100 years. Summary of data extracted from Table 4. Alaska and Hawaii has not been examined [Colour figure can be viewed at wileyonlinelibrary.com]

parameter of interest is the time trend coefficient to check if there has been an increase in the temperatures. However, instead of imposing that the errors in the detrending regression were $I(0)$ as is standard in the literature, we suppose that they are fractionally integrated or $I(d)$ where the differencing parameter d is also estimated from the data. In doing so we get more precise estimates of the time trend coefficients, especially noting that we find evidence of long memory ($d > 0$) behaviour in all the examined states. Except for 10 states, (namely, Alabama, Arkansas, Georgia, Kentucky, Louisiana, Mississippi, Missouri, Oklahoma, Tennessee and West Virginia), we find evidence of an increase in the temperature anomalies (with respect to 1901–2000), with the values increasing between 1.5°C and 2.5°C over 100 years in the majority of the cases. The highest values are observed in the cases of New Jersey and Rhode Island with increases of around 2.9°C every 100 years.

The evidence in favour of long memory behaviour in US temperature data is relevant in the sense that we should expect more likely this long memory pattern in aggregated data rather than in data based on specific locations.[8] Also, in an hydrological context, fractional integration is justified in terms of aggregation of heterogeneous spatial patterns (Mudelsee, 2007); thus, in general, higher degrees of integration must be observed in aggregated data than in local ones. Nevertheless, we obtain evidence of long memory in all the 48 series examined, and taking into account this fact, the estimation of the time trend coefficients differ, in some cases substantially, from those obtained based on $I(0)$ errors, which has been one of the main contribution of this work. Note, however, that we have not make any hypothesis about why this increase in the temperatures has taken place especially in the North-East regions, while no warming was observed in the South-East regions. This is something that has been left for future works.

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ENDNOTES

[1] For instance, if x_t is AR(1), we can use the Prais and Winsten (1954) transformation, to obtain a t-statistic, which converges in distribution to a $N(0,1)$ random variable. However, as noted by Park and Mitchell (1980) and Woodward and Grey (1993), significant size distortions may appear in the test statistic when the coefficient is close to 1. If the correlation structure of u_t is higher than AR(1), the OLS estimates of β_1 is asymptotically equivalent to GLS in the case of known correlation (see, for example, Grenander and Rosenblatt, 1957).

[2] An $I(0)$ process is defined as a covariance stationary process where the infinite sum of the autocovariances is finite. Thus, it includes the stationary and invertible ARMA processes.

[3] Long memory is also known as long range dependence, strong dependence, global dependence or the Hurst phenomena (see, Rea *et al.*, 2011; Graves et al., 2017).

[4] Using other methods like the time domain maximum likelihood methods in Sowell (1992) and Beran (1995) produced almost identical results.

[5] Note that seasonal fractional integration take place when $(1 - B^{12})$ is replaced by $(1 - B^{12})^d$. See, Gil-Alana and Robinson (2001). However, that model is not considered in this work.

[6] This non-parametric approach accommodates extremely well in the context of fractional integration (see, Gil-Alana, 2004).

[7] This can be explained by the fact that the seasonal differencing polynomial $(1 - B^{12})$ can be expressed as $(1 - B)\ S(B)$ where $S(B) = (1 + B + B^2 + \ldots + B^{11})$, and thus, the first of this polynomials refers to the long run or zero frequency.

[8] Note that long memory processes were initially justified in terms of aggregation of heterogeneous AR process by Robinson (1978) and Granger (1980). Other authors also using aggregation as a theoretical argument for long memory are Taqqu *et al*. (1997), Chambers (1998), Lippi and Zaffaroni (1999), Parke (1999), etc.

ORCID

Luis A. Gil-Alana https://orcid.org/0000-0002-5760-3123

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